

MEAN STACK



MERN is a popular JavaScript stack used for building modern web applications. MERN gets its name from 4 technologies that make up the stack which are MongoDB, Express, React, Node.

Advanced Js

How Javascript Works

- hoisting
- strict mode
- closure
- synchronous and asynchronous programming

Event propagation

- bubbling
- capturing

Event loop

function curring

ajax and json

calling api using ajax

callback hell

promise

async await

fetch

modules

version

Typescript

- Why Typescript
- Basic Types
- Class and Interfaces

Angular

INTRODUCTION TO ANGULAR

- What is Angular
- What is SPA
- Comparison between different versions
- Advantages and Disadvantages
- Key features

ENVIRONMENTAL SETUP

- Node /NPM
- Angular CLI
- Setup Project
- Directory Structure
- Code Editors
- How Angular Application Boot

MODULES

- What is Module
- Decorator
- NgModule
- Core Module
- Bootstrapping
- Shared Modules

COMPONENTS

- Introduction to Components
- Component Architecture Patterns
- Metadata
- State and Behaviors

TEMPLATES

- Template Expression or Interpolation
- Data Binding (Event, Property, Two Way)
- *ngIf, *ngFor, *ngSwitch
- Built-in Structural and Attributes Directive

PIPES

- Built-in Pipes
- Custom Pipes

Routing

- Routes, RouterModule, routerLink
- Child Routing
- Params
- Guards

CUSTOM DIRECTIVES

- Structured Directives
- Attribute Directives

VALIDATIONS

- Template Driven
- Reactive Form
- Custom Validators

ASYNCHRONOUS OPERATIONS

- Observable vs Promise
- Network Call
- HTTP Request | Response

SERVICES

- Introduction to Services
- Building a Service
- Dependency Injections
- Injectors and Providers

LIFECYCLE HOOKS

- Component LifeCycle
- All LifeCycle Hooks

LOCAL DATA STORAGE

- Local Storage
- Session Storage
- Cookies

COMMUNICATION BETWEEN COMPONENTS

- @Input()
- @Output() with EventEmitter
- @ViewChild()
- Shared Service
- Subject

SERVER-SIDE RENDERING

- Introduction to Server-Side Rendering
- Angular Universal

Angular Testing

- Introduction to Testing
- Unit Testing with Karma and Jasmine

LAZY LOADING

UI COMPONENTS

- Angular Material
- PrimeNG

Third Party Modules

- Social Login
- Pagination
- Search
- Filter
- JWT Token
- File Upload
- Many More

Angular Application Deployment

- Build Application
- Deploy in Heroku

Node Js

Introduction to Node.js

- What is node.js
- A word on event loop
- Event driven architecture
- Blocking vs non-blocking code
- Node Process Model

- Setup node.js
- Node REPL
- Debugging node.js app

Modules in JS

- Understanding exports and require
- Creating modules
- Importing modules
- Quick tour on NPM
- Installing 3rd party modules

Events & streams

- Significance of Events
- Event Emitter class
- Emitting and listening to events
- Types of streams
- Working with streams
- Composing streams using pipe

Working with Process and File System

- Process object
- Handling exceptions at process level
- Listening to process events
- Working with file system using fs module
- Creating, Copying, Deleting files
- Creating, Copying, Deleting directories
- Watching for file changes

HTTP

- Building a Web Server
- HTTP Request Methods, Headers
- Response Codes and Headers

Express Js

Expressjs Introduction & Setup

- Building a Web Server
- HTTP Request Methods, Headers
- Response Codes and Headers
- Serving static files with express
- Express application generator

Routing

- Route Methods
- Route Paths
- Parameterized Routes
- Route Handlers
- Express Router

Middleware

- Types of middleware

1. Application
 2. Router
 3. Error
 4. Built-in
 5. Third-party
- Error middleware
 - Body parser, cors, cookie parser, session management

Template Engines

- What are template engines
- EJS
- Jade
- Handlebars

Expressjs security

1. Authentication
 2. JWT
 3. Securing routes
- Debugging in Express
 - Realtime programming with Socket.io

Mongo DB

Introduction to MongoDB

- Installing MongoDB
- The current SQL/NoSQL landscape
- Document-oriented vs. other types of storage
- Mongo's featureset
- Common use-cases
- MongoDB databases
- MongoDB Collections
- MongoDB Documents

CRUD Operations in MongoDB

- Creating documents
1. insert()
 2. update()
 3. save()
- Querying documents
4. find()
5. Working with equality
 6. Query operators
 7. Building complex queries
- Updating documents
 - Deleting documents
 - Limit
 - Order By

Introduction to Mongoose

- Word on ORM/ODM
- Installing mongoose
- Connecting to MongoDB from mongoose

Core concepts of Mongoose

- Understanding mongoose schemas and datatypes
- Working with Models
- Using modifiers in schema
- Using virtual fields
- Optimizing query performance by enabling indexes

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